# CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION

**Cambridge IGCSE™ & O Level Computer Science (0478 / 2210)**

**TOPIC 1.2: TEXT, SOUND AND IMAGES • OFFICIAL TEACHER MARK SCHEME**

**Syllabus: 0478 / 2210 Topic 1.2** | **Maximum Mark: 50 Marks** | **Complete Step-by-Step Worked Solutions**

## PART A: TEXT REPRESENTATION & CHARACTER SETS SOLUTIONS

### Question 1 [5 marks]

**(a) Character Set Definition:** [2 marks]
• A defined list of characters/symbols recognized and encoded into binary numbers by a computer system [1].
• Each character is represented by a unique binary code [1].

**(b) ASCII vs Unicode Differences:** [2 marks, 1 mark per point]
• ASCII uses 7 or 8 bits (128 to 256 characters) whereas Unicode uses 16 or 32 bits (over 1 million characters) [1].
• ASCII only supports English/Latin characters whereas Unicode supports all global languages, math symbols, and emojis [1].

**(c) Why Unicode is Preferred:** [1 mark]
• Allows international communication and multilingual text processing across the globe.

### Question 2 [4 marks]

**(a) File size in Bytes:** [1 mark]
• 4,096 characters × 1 Byte per character = **4,096 Bytes** [1]

**(b) File size in Kibibytes (KiB):** [3 marks]
• Step 1: Total Bytes = 4,096 Bytes [1]
• Step 2: Division by 1024 → 4,096 ÷ 1,024 [1]
• Final Answer: **4 KiB** [1]

## PART B: SOUND REPRESENTATION & SAMPLING SOLUTIONS

### Question 3 [6 marks]

**(a) Sound Sampling Process:** [3 marks]
• Continuous analogue sound waves are captured by a microphone [1].
• The wave amplitude is measured/sampled at regular time intervals [1].
• Each sampled amplitude is converted into a binary number and stored [1].

**(b) Hardware Device: ADC (Analogue-to-Digital Converter)** [1 mark]

**(c) Definitions:** [2 marks]
• **Sample Rate:** Number of sound samples taken per second (in Hz) [1].
• **Sample Resolution:** Number of bits allocated to store each sound sample [1].

### Question 4 [5 marks]

**(a) Total size in bits:** [2 marks]
• Formula: Sample Rate × Bit Depth × Time × Channels
• Calculation: 44,100 × 16 × 20 × 2 = **28,224,000 bits** [2]

**(b) Uncompressed size in MB:** [3 marks]
• Convert bits to Bytes: 28,224,000 ÷ 8 = 3,528,000 Bytes [1]
• Convert Bytes to MB: 3,528,000 ÷ 1,000,000 = **3.528 MB** [2]
• *(Alternative in MiB: 3,528,000 ÷ 1,048,576 ≈ 3.36 MiB is also fully credited)*

## PART C: IMAGE REPRESENTATION SOLUTIONS

### Question 5 [6 marks]

**(a) Image Definitions:** [3 marks]
• **Pixel:** The smallest single picture element of a bitmap image [1].
• **Resolution:** Total pixel dimensions of the image (Width × Height) [1].
• **Colour Depth:** Number of bits used to represent the colour of each pixel [1].

**(b) Unique Colours:** [2 marks]
• (i) 4 bits: 2^4 = **16 colours** [1] • (ii) 8 bits: 2^8 = **256 colours** [1]

**(c) Metadata Example:** Image resolution, colour depth, date created, camera model, GPS tags [1 mark].

### Question 6 [5 marks]

**(a) Total Image Size in Bytes:** [2 marks]
• Total bits = 2000 × 1500 × 24 = 72,000,000 bits [1]
• Total Bytes = 72,000,000 ÷ 8 = **9,000,000 Bytes** (or 9 MB) [1]

**(b) Image Size in Mebibytes (MiB):** [3 marks]
• Calculation: 9,000,000 ÷ 1,048,576 = **8.58 MiB** (allow 8.58 to 8.6 MiB) [3 marks for correct working and answer].

## PART D: DATA COMPRESSION SOLUTIONS

### Question 7 [6 marks]

**(a) Table Ticking:** [3 marks]
• Social media photograph (JPEG) → **Lossy Compression** [1]
• Program executable (EXE) → **Lossless Compression** [1]
• Music track for streaming (MP3) → **Lossy Compression** [1]

**(b) Why Lossless is Required for Code/Text:** [2 marks]
• Deleting even a single character or byte from program code causes syntax errors and crashes [1].
• Lossless guarantees 100% exact bit-for-bit reconstruction of the original file [1].

**(c) Benefit of Compression for Video:** Faster streaming without buffering / consumes less user bandwidth [1 mark].

### Question 8 [5 marks]

**(a) RLE Description:** [2 marks]
• A lossless algorithm that identifies consecutive repeated runs of identical data [1].
• Replaces the run with a count and single data value pair [count, value] [1].

**(b) RLE Solution:** [3 marks]
• `W W W W W W` → 6W [1]
• `B B B` → 3B [0.5]
• `W W W W` → 4W [0.5]
• `B B B` → 3B [1]
• Full Answer: **6W 3B 4W 3B** (or `06 W 03 B 04 W 03 B`) [3 marks].

## PART E: MULTIPLE CHOICE DRILL SOLUTIONS

### Question 9 [8 marks]

1. Standard ASCII bits → **a) 7 bits** [2]
2. Sample Rate unit → **b) Hertz (Hz)** [2]
3. 8-bit colour depth unique colours → **b) 256 colours** [2]
4. Lossless compression example → **c) PNG / RLE** [2]